Summary:

This feature handles the creation and access of books and authors. Books and authors are related to each other in a Many-to-One relationship and stored in the PostgreSQL database under the 'book' and 'author' columns respectively. The primary key for books within the database is the book's ISBN number. The primary key for authors is an internal authorId which increments with each author added.

API endpoints:

There are 4 main endpoints added with this feature:

1. POST REQUEST: /authors/
   This endpoint is used to add authors to the database. To submit an author a JSON must be uploaded to this endpoint with the following structure:
   ```
   {
     "firstName": "",
     "lastName": "",
     "biography": ""
     "publisher": {
       "publisher_name": ""
     }
   }
   ```
   Note: This endpoint will also create the publisher during the author creation process if it does not exist and associate them.

2. POST REQUEST: /books/
   This endpoint is used to add books to the database. To submit a book a JSON must be uploaded to this endpoint with the following structure:
   ```
   {
     "isbn": "",
     "bookName": "",
     "bookDescription": "",
     "bookPrice":0.0,
     "bookYearPublished":0000,
     "bookCopiesSold":0,
     "avgRating":0.0,
     "bookAuthor": {
       "authorId": 0
     },
     "bookGenre": {
       "genre": ""
     },
     "bookPublisher": {
       "publisher_name": ""
     }
   }
   ```

Note: This endpoint will create the genre if it does not exist, but it will NOT create an author or publisher. Please ensure you have created an author and publisher with the endpoint above and use what authorId it has been assigned to create a book associated with that author.

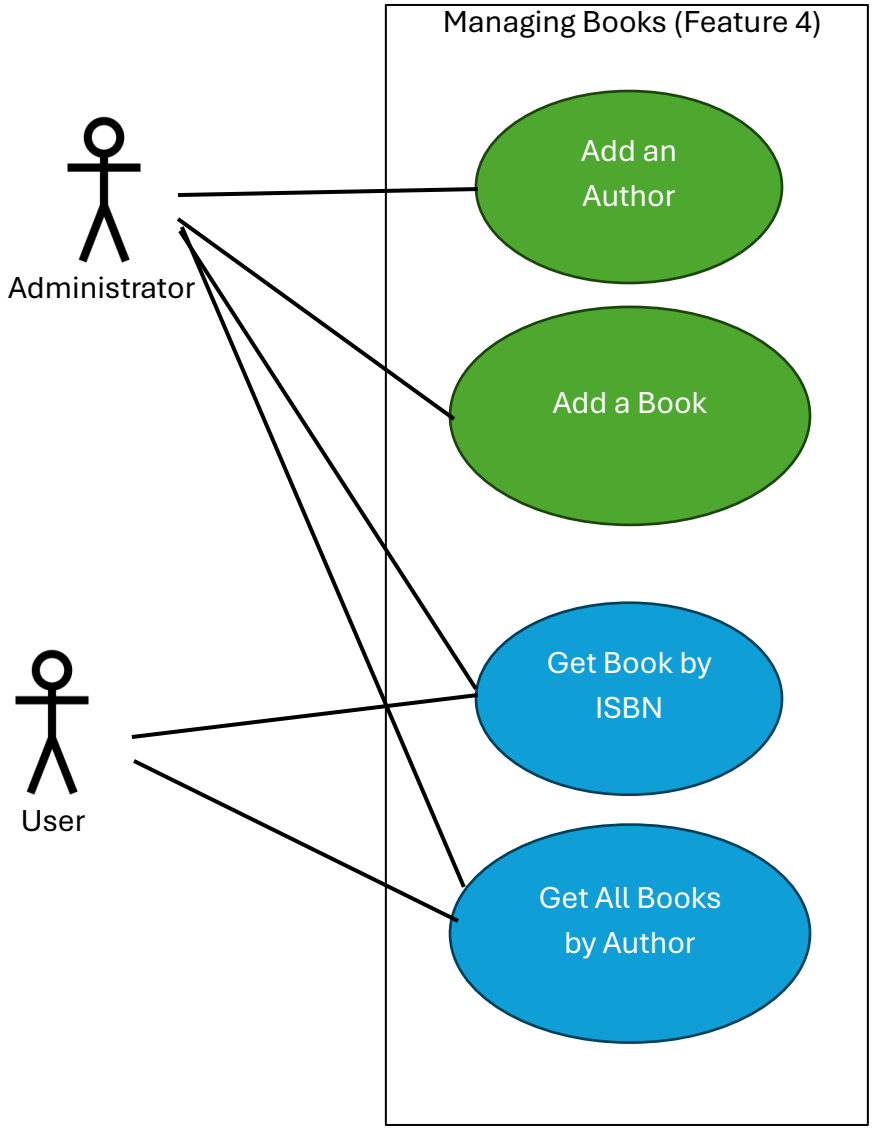
3. GET REQUEST: /books/{isbn}
   This endpoint will retrieve the book from the database with the matching ISBN. No input needs to be provided, simple place the ISBN at the end of the link.

4. GET REQUEST: /authors/{authorId}/books
   This endpoint will receive all the books that have been associated with an author. Similarly to the get request above, the only input required is to put the internal id of the author whose books you'd like to retrieve in the link. If you do not know an author's id, check their location in the database as their id is their key within the database.

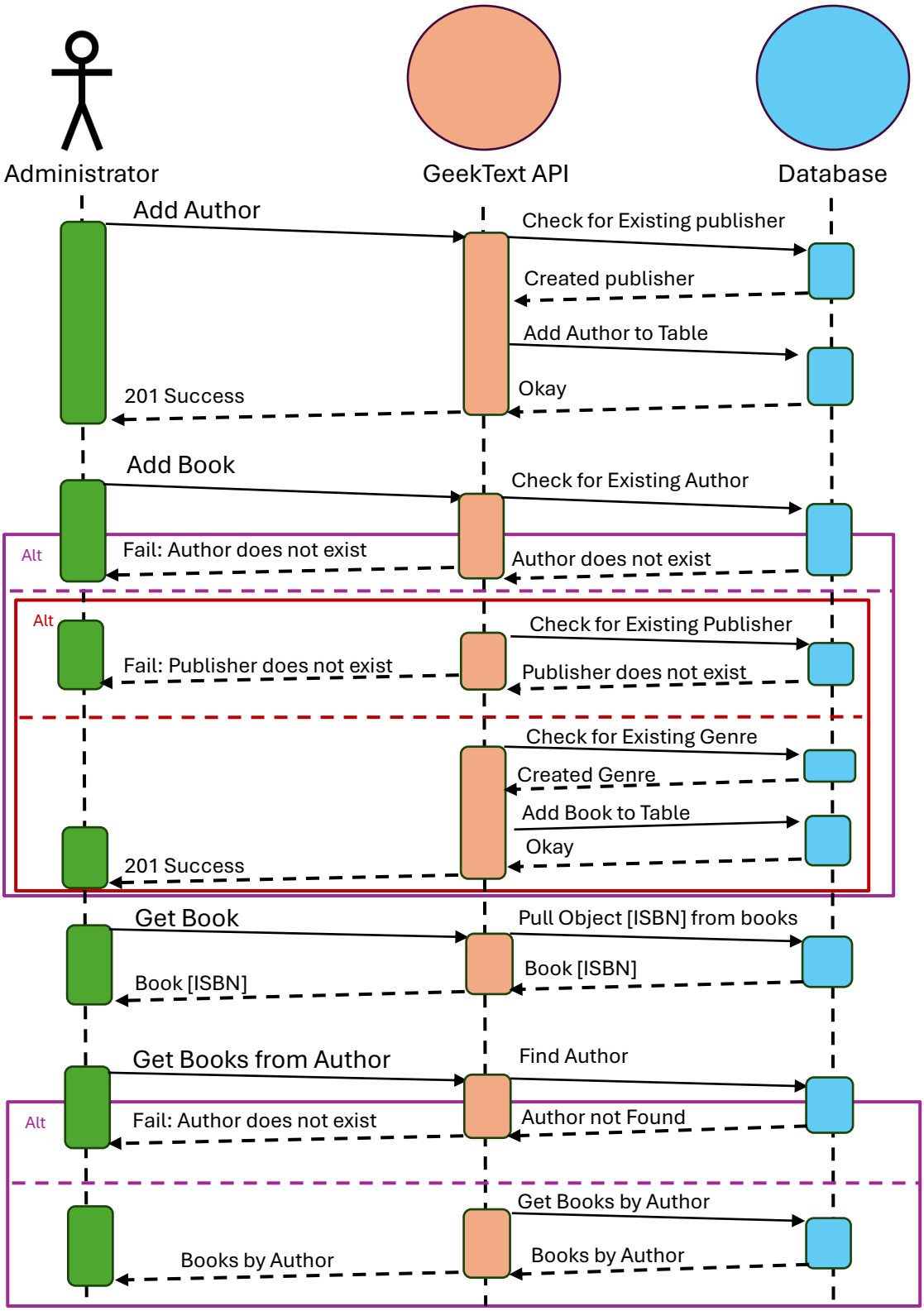
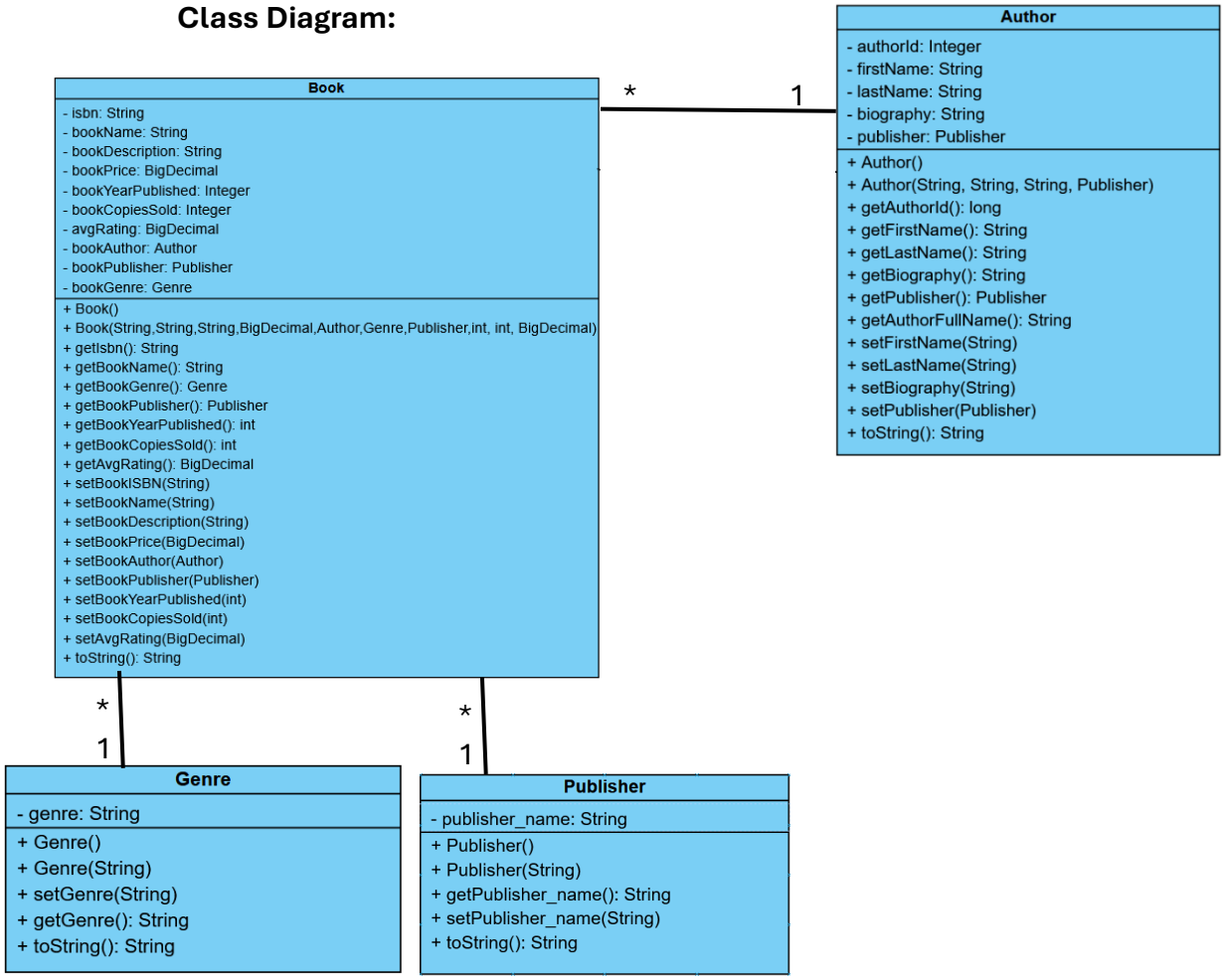
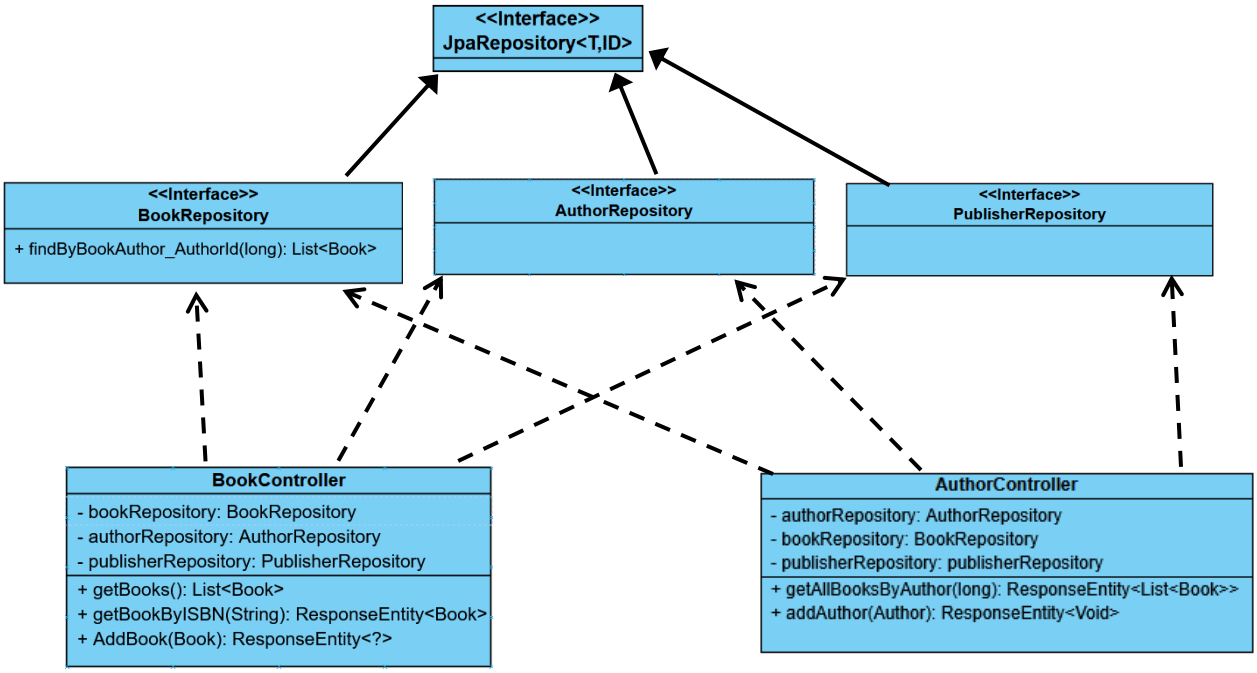
Additionally, although it is not meant for day-to-day use, you can use a get request on the /books/ endpoint to retrieve all the books within the database. Again, this endpoint is available for testing purposes and should not be needed for day-to-day operations.

Included in the pages below are the class, sequence, and use case UML diagrams for further reference.

Use Case Diagram:



Sequence Diagram:



Class Diagram: